necessarily be moving to the right at the same speed—we cannot predict the actual numerical value of the speed with which the particles move. Instead, the best we can do is use the laws of quantum physics to predict the probability that any particular speed is the one attained. Similarly, while we can say with certainty that there is a definite relationship between the positions of the particles—if one is measured at a given moment and found to be at some location, the other necessarily is located the same distance from the starting point but in the opposite direction—we cannot predict with certainty the actual location of either particle. Instead, the best we can do is predict the probability that one of the particles is at any chosen location. Thus, while quantum mechanics does not give definitive answers regarding particle speeds or positions, it does, in certain situations, give definitive statements regarding the *relationships* between the particle speeds and positions.

Einstein, Podolsky, and Rosen sought to exploit these relationships to show that each of the particles actually has a definite position and a definite velocity at every given instant of time. Here's how: imagine you measure the position of the right-moving particle and in this way learn, indirectly, the position of the left-moving particle. EPR argued that since you have done nothing, absolutely nothing, to the left-moving particle, it must have *had* this position, and all you have done is determine it, albeit indirectly. They then cleverly pointed out that you could have chosen instead to measure the right-moving particle's velocity. In that case you would have, indirectly, determined the velocity of the left-moving particle without at all disturbing it. Again, EPR argued that since you would have done nothing, absolutely nothing, to the left-moving particle, it must have *had* this velocity, and all you would have done is determine it. Putting both together—the measurement that you did and the measurement that you *could* have done—EPR concluded that the left-moving particle has a definite position and a definite velocity at any given moment.

As this is subtle and crucial, let me say it again. EPR reasoned that nothing in your act of measuring the right-moving particle could possibly have any effect on the left-moving particle, because they are separate and distant entities. The left-moving particle is totally oblivious to what you have done or could have done to the right-moving particle. The particles might be meters, kilometers, or light-years apart when you do your